EUROCONTROL

Airport Operator Data Flow – Data Specification

Operational ANS Performance Data Items

Summary

This specification defines and documents the requirements for the Airport Operator Data Flow.

This initial version is prepared for consultation and in support of the on-going development of a web-interface for the Airport Operator Data Flow.

The approved version will be quality controlled under the Quality Assurance System established for the Airport Operator Data Flow and maintained by the EUROCONTROL Performance Review Unit.

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Abstract

This specification defines and documents the data requirements for the Airport Operator Data Flow in support of the regular operational ANS performance monitoring and reporting under the Single European Sky and EUROCONTROL Performance Scheme.

Keywords

Airport Operator Data Flow	Data Specification	Operational Data Item	Record Format

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1. Introduction

1.1 General

This specification defines and documents the data-technical requirements for the Airport Operator Data Flow. The Airport Operator Data Flow covers the processes, procedures, and data for the purpose of operational performance monitoring and reporting of air navigation services at airports.

This document provides material in form of specific requirements which need to be met by reporting entities when providing data for the purpose of operational performance monitoring and reporting.

This document ensures compliance with the data requirement of the Single European Sky and EUROCONTROL Performance Scheme.

1.2 Purpose of the document

This document

- defines and establishes the processes and procedures for the regular operational performance data reporting;
- specifies the format of the respective data files, records and operational data items; and
- identifies the requirements on the performance data in terms of processing and quality assurance.

The specification is established under the Quality Assurance System of the EUROCONTROL Performance Review Unit for the Airport Operator Data Flow and serves for communication with the involved stakeholders.

Compliance with this specification ensures timely and robust data for the purpose of operational performance monitoring and reporting under the aforementioned performance schemes.

Stakeholders are required to demonstrate compliance with this specification before regularly submitting their operational performance data. Following the demonstration of the operational readiness, stakeholders shall ensure compliance with this specification as part of the regular monthly performance reporting.

1.3 Scope

The scope of this document addresses the Airport Operator Data Flow (APDF). This specification acknowledges quality assurance and data quality requirements. It introduces requirements in the form of provisions which place controls on the processes applied to the performance data, in particular during the submission and initial processing phases. Down-stream processes of the APDF are governed by the quality processes established for the APDF.

Through this approach, the quality of performance data is assured by demonstrating that the processes applied give the required degree of assurance that the data will not be adversely affected. Establishing and maintaining data with the required level of integrity is only part of the solution. If submitted data is not originated correctly, the resultant erroneous or ill-formed data will be processed. In essence, the system becomes one of “rubbish in – rubbish out” with a high level of assurance that erroneous or ill-formed data will not be altered. Hence, compliance with this standard ensures a robust set of data for the calculation of performance metrics and performance indicators. These are strongly dependent on the integrity and availability of the specified operational performance data items. Coherency of the metrics and indicators can only be assured, if the required performance data is provided with a high degree of quality and completeness.

To address this, this specification includes provisions which are specifically intended to be met by the respective reporting entities.

The data flow is established for the purpose of air navigation performance monitoring for airports with an average of more than 70.000 flights per annum.

National authorities may further decide in response to regulatory requirements or on a voluntary basis to enlist airports to comply with this specification irrespective of this threshold.

Airports averaging below the aforementioned threshold or not nominated by their national authorities may decide to join the operational performance monitoring and comply with this specification.

In all cases, the mandatory requirements in this specification must be met in order to be considered compliant with the performance data reporting requirements of the airport operator data flow (APDF).

1.4 Conventions and Requirement Characterisation

This specification contains the set of requirements for assuring quality of the performance monitoring and reporting during the submission and initial processing phases of the APDF.

The following conventions apply:

- "**shall**" indicates a statement of the specification, the compliance with which is mandatory to achieve compliance with this specification; it indicates a requirement which must be satisfied by all parties claiming conformity to this specification. Such requirements shall be testable and their implementation auditable.
- **"should"** indicates a statement of the specification, the compliance with which is strongly advised to achieve compliance with this specification; it indicates a requirement which is recommended to be satisfied to achieve the best possible implementation of this specification. Generally, these requirements shall be testable and their implementation audible.

Each requirement in this specification is preceded by an identifier to allow for better traceability. The requirement identifier applies the following naming convention:

APDF-[Pn]-[nnn]-[C]

where

- APDF refers to the Airport Operator Data Flow (this specification);
- [Pn] decodes the respective procedure or process category to which the requirement applies;
- [nnn] is a numeric identifier; and
- [C] specifies whether the requirement is mandatory (M), recommended (R), or highly desirable (HD).

1.5 Review and Refinement of Data Specification

This data specification is a controlled document under the Quality Assurance Framework established by the EUROCONTROL Performance Unit for operational performance data reporting.

The specification and lessons learnt from day-to-day application of the APDF processes and quality procedures may result in the integration of further requirements and considerations to strengthen the robustness of the data.

Such changes will be coordinated with the reporting entities subject to this specification and their implementation will be subject to an appropriate transition plan.

1.6 Definitions and Acronyms

Term	Description
APDF	Airport Operator Data Flow The set of processes and procedures comprising the data collection and processing of operational performance data for the purpose of regular performance monitoring and reporting under the Single European Sky and EUROCONTROL Performance Scheme.
CODA	Central Office for Delay Analysis
EATM	EUROCONTROL Air Traffic Management Programme
IFPS	Initial Flight Plan Processing System
(K)PI	(Key) Performance Indicator Performance indicator refers to an indicator used for the purpose of performance monitoring, benchmarking and reviewing. Key performance indicator refers to an indicator used for the setting of performance targets.
NM	Network Manager
ODI	Operational Data Item The operational data item reflects the data to be reported under this specification including the associated quality requirements on the data.
PRB	Performance Review Body
PRC	Performance Review Commission
PRISME	Pan-European Repository of Information Supporting the Management of EATM
PRU	Performance Review Unit
Reporting Entity	The entity (e.g. function, office) identified being responsible for the regular provision of the data specified in this document.
SES	Single European Sky The single European sky initiative of the European Commission.

Table 1 - Definitions and Acronyms

2. Airport Operator Data Flow

This Chapter summarises the process, procedures and applicable policies of the APDF. Conceptually, the APDF covers all processes and procedures concerning the collection and processing of performance related data described in this specification, including the calculation of related performance measures.

2.1 Data Flow Process and Actors

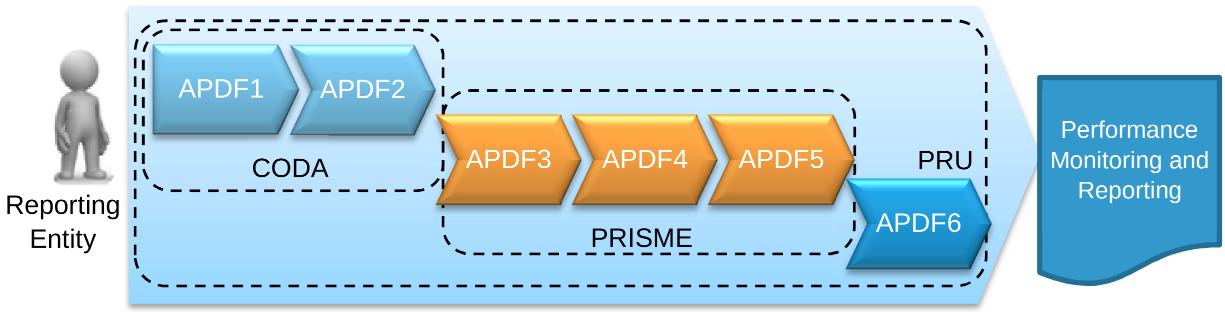

Figure 1 – Conceptualisation of Airport Operator Data Flow

The APDF relates to the set of processes and procedures established for the collection and processing of operational performance monitoring and reporting data. These data are required to calculate the operational ANS performance indicators.

The scope of the APDF comprises the following sub-processes:

- APDF1 – data submission and collection
- APDF2 – data quality assessment
- APDF3 – extract and load data
- APDF4 – merge and load data
- APDF5 – pre-computation of performance parameters
- APDF6 – calculation of performance indicators

Figure 1 presents the conceptualisation of the APDF and the involved actors. The following principal roles can be identified:

Actor	Description
Reporting Entity	The entity (e.g. function, office) identified being responsible for the regular provision of the data specified in this document.
Central Office for Delay Analysis (CODA)	CODA is responsible for the data collection and initial processing of data submitted for the purpose of performance monitoring and review.
Pan-European Repository of Information Supporting the Management of EATM (PRISME)	PRISME is responsible for the day-to-day management of the EUROCONTROL data warehouse processing of the performance data.
Performance Review Unit (PRU)	PRU assumes responsibility for the APDF and is responsible for the timely publication of performance monitoring and review deliverables.

Table 2 – Airport Operator Data Flow Actors

2.2 Quality Assurance Approach

Quality assurance is focused on providing confidence that quality requirements will be fulfilled. In other words, it pertains to all those planned and systematic actions necessary to provide adequate confidence that a product or service will satisfy the requirements for quality. This is a fundamental shift in concept from the reactive downstream approach of quality control by means of detection, to a proactive upstream approach that controls and manages the upstream activities to prevent problems from arising.

This specification follows a quality assurance approach and aims to address

- data quality; and
- process quality of the APDF.

Data quality refers to the degree of excellence exhibited by the data comprising the state of completeness, validity, consistency, timeliness, and accuracy that makes the data appropriate for the purpose of performance monitoring and reporting.

For the purpose of quality assurance the following data quality dimensions have been identified:

Dimension	Description
Format	The process of translating, arranging, packaging and processing a selected set of data for submission under this specification. As a result, this specification defines a data structure (i.e. format) that fulfils the characteristics of data quality.
Syntax	The set and sequence of permissible symbols forming an operational data item.
Completeness	The degree of confidence that all of the data, needed to support the intended use, has been provided.
Validity	The degree of confidence that data are collected and used in compliance with internal and external requirements, to ensure consistency and that they appropriately reflect what they are intended to measure.
Consistency and Comparability	The coherence of two or more data outputs refers to the degree to which processes by which data / information were generated used the same concepts - classifications, definitions, and target populations – and harmonised methods. Coherent outputs have the potential to be validly combined and used jointly. Comparability is a special case of coherence and addresses data / information outputs referring to the same data items and the aim of combining them is to make comparisons over time, or across regions, or across other domains.
Timeliness	The degree of confidence that the data are applicable to the period of its use, are promptly collected and reported within the timeframes specified in this specification.
Accuracy	The degree of conformance between the estimated or measured value and its true value.

Table 3 – Data Quality Dimensions

Process quality is linked to the APDF processes and includes a balance between sound methodology, appropriate procedures, non-excessive burden on respondents and cost effectiveness.

Documentation on the established APDF processes and procedures is available through the DANSAP One Sky Team.

2.3 Monthly Verification and Validation

The APDF1 and ADPF2 sub-processes and associated responsibilities have been defined above. Figure 2 presents the high-level process for the submission, verification, and validation of the operational performance data submitted by the reporting entity in compliance with this specification. This high-level process is mapped against the general reporting time line.

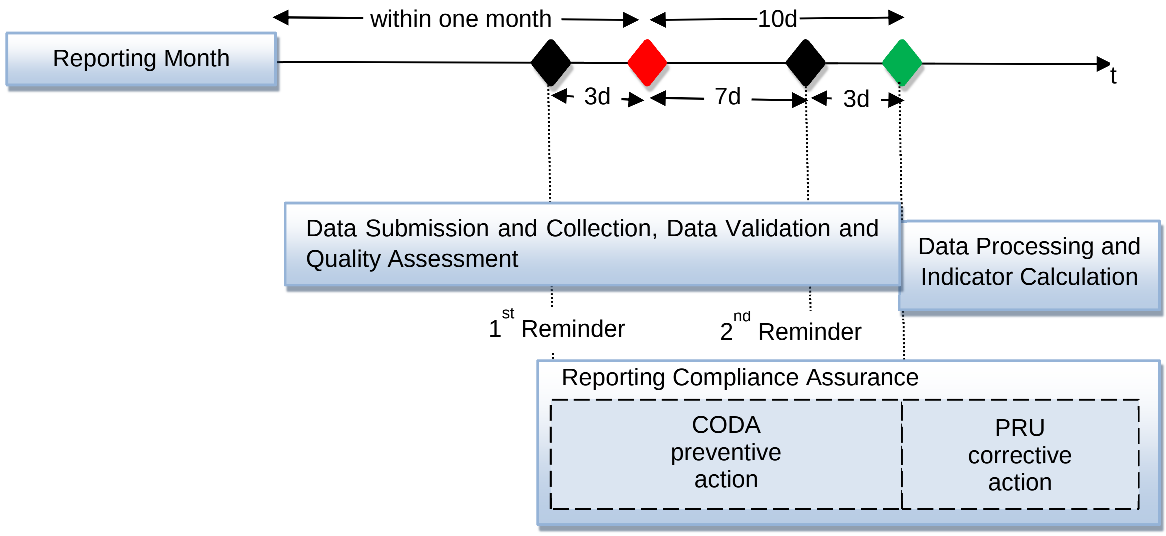

Figure 2 – APDF Data Submission, Verification, and Validation

The monthly verification and validation is designed to support the regular reporting requirements and to ensure a consistent timeline of performance monitoring and reporting. In particular, performance monitoring results (i.e. performance indicators) shall be made available as soon as possible. For that purpose, it is foreseen that the further processing of performance data, i.e. ADPF3 through ADPF6, is effected following the 10th day after the end of the reporting cycle.

To achieve this aim, the following principles apply:

APDF-DSC-010-M The reporting entity **shall** provide the data within one month of the end of the reporting month.

APDF-DSC-020-R To allow for a shorter reporting cycle, the reporting entity **should** provide the data **before the 20th day of the month** following the reporting month.

As part of the data collection process, preventive action shall be taken whenever a reporting entity may run or runs late to comply with the reporting requirement. If the reporting entity fails to provide the data within a defined time period, corrective action shall be initiated to ensure timely reporting or, if required, to initiate the escalation process and identify an appropriate remedial action plan.

In particular,

APDF-DSC-030-M CODA **shall** send a first reminder to the reporting entity 3 days before the end of the month following the reporting month.

APDF-DSC-040-M If the reporting entity fails to provide the data before the 7th day of the following month, CODA **shall** issue a 2nd reminder with a deadline of three further days.

APDF-DSC-050-M PRU **shall** initiate and coordinate corrective action by advising the reporting entity about the logged non-compliance and inquire – in collaboration with the respective authority – the provision of the data or an appropriate remedial action plan.

2.4 Non-Conformance Handling

The vision of the timeline described above is that data related or data processing related imperfections are remedied on the basis of a thorough follow-up of non-conformances. This typically results in remedial actions (e.g. trouble shooting, remedial action coordination with reporting entity).

Completed remedial actions will be re-integrated with next suitable timeline iteration. For example, if the PIs for an airport cannot be calculated, this airport will not be included in the update of the dashboard. Upon reception of the data, the associated performance measures will be published with the next update of the dashboard.

Non-conformances are identified by the breach of the quality criteria of this specification. Associated quality controls and business rules are tailored for each process step. Major non-conformances result in a stop of the further processing of the data as the overall aim (robustness and quality of performance data) is no longer ensured. For APDF1 and APDF2, this specification lists the respective quality criteria and business rules established.

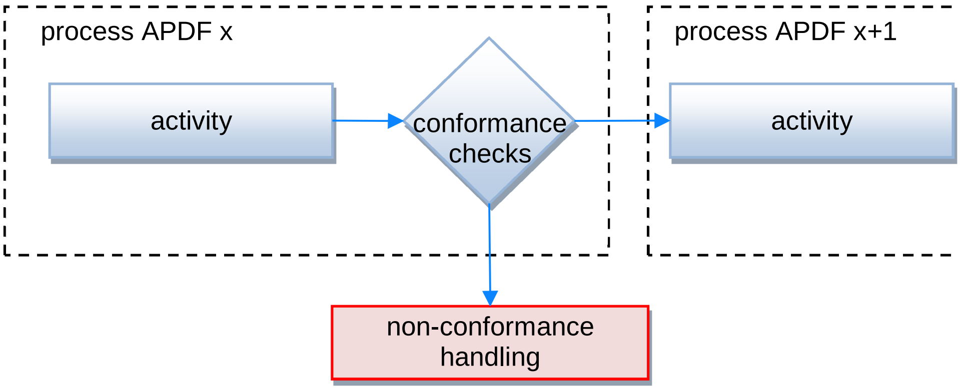

Figure 3 – Exception Handling

APDF-NCH-010-M PRU and CODA **shall** inform the reporting entity about the identified non-conformances.

APDF-NCH-020-M Reporting entities **shall** address non-conformances by re-submitting data in compliance with this specification by no later than the subsequent reporting cycle.

Remedial action requiring more than a month (i.e. next reporting cycle) shall be coordinated with the PRU.

APDF-NCH-025-R Reporting entities **should** re-submit the operational performance data within 14 days of notification.

APDF-NCH-030-M PRU **shall** withdraw the compliance status of a reporting entity in case of persistent non-conformances (i.e. two consecutive reporting cycles) and inquire the submission of a remedial action plan by the reporting entity.

The launch of the inquiry of remedial action plans shall be notified to the sponsor bodies of the performance scheme (i.e. SES Performance Scheme: PRB/EC, EUROCONTROL Performance Scheme: PRC).

APDF-NCH-040-M PRU **shall** launch the escalation procedure with the appropriate national authority for airports subject to the SES Performance Scheme and the respective rules of the regulation.

Note: escalation procedure refers to the coordination with the national authority that shall ensure compliance of the reporting entity with the data reporting requirements and the associated quality of the data.

3. Normative Specification for the Airport Operator Data Flow

3.1 Airport Operator Data Flow – Data Policy

Performance monitoring and review is strongly dependent on appropriate and robust data. For that purpose, PRU established a quality assurance framework to ensure the consistent processing of data.

This specification defines and documents the data-technical requirements for the Airport Operator Data Flow. The Airport Operator Data Flow covers the processes, procedures, and data for the purpose of operational performance monitoring and reporting.

The principal data quality policy for the APDF comprises the

- appropriate data collection, processing, analysis and reporting of accurate, reliable, and consistent performance data;
- establishment of an assurance framework to ensure sufficient action is being taken to meet data quality requirements; and
- compliance with audit standards and requirements.

The policy is accompanied by a robust set of processes and procedures with clearly identified actions, responsibilities, and timescales to support continual improvement and pro-active remedial action planning to drive data quality of the regular performance monitoring and reporting.

APDF-POL-010-M Data provided under this specification by the reporting entity **shall** comply with all the requirements set out for the respective operational data items.

In particular, the

- message syntactical and formatting requirements;
- record related requirements; and
- operational data item specific requirements

shall be met.

APDF-POL-020-M Files / data submitted which contain missing or imperfect data required to calculate the performance measures **shall** be rejected in accordance with the non-conformance thresholds and rules specified in this specification.

APDF-POL-030-M Rejected data **shall** be reviewed by the reporting entity and re-submitted in compliance with this specification and the timeframes mentioned in Section 2.4.

3.2 File Format and Syntax

For the purpose of performance monitoring and review, this specification sets out the format and syntax of files, records, and operational data items (ODI) required for the departures and arrivals at an aerodrome (c.f. Figure 4). A set of ODIs reflect the individual flight related information (i.e. 'record'). Dependent on the type of file and record, a field specifies the physical location of the required ODI. Detailed specification requirements for each ODI are given in Chapter 4. All records for the respective reporting month are submitted through a separate departure and arrival file.

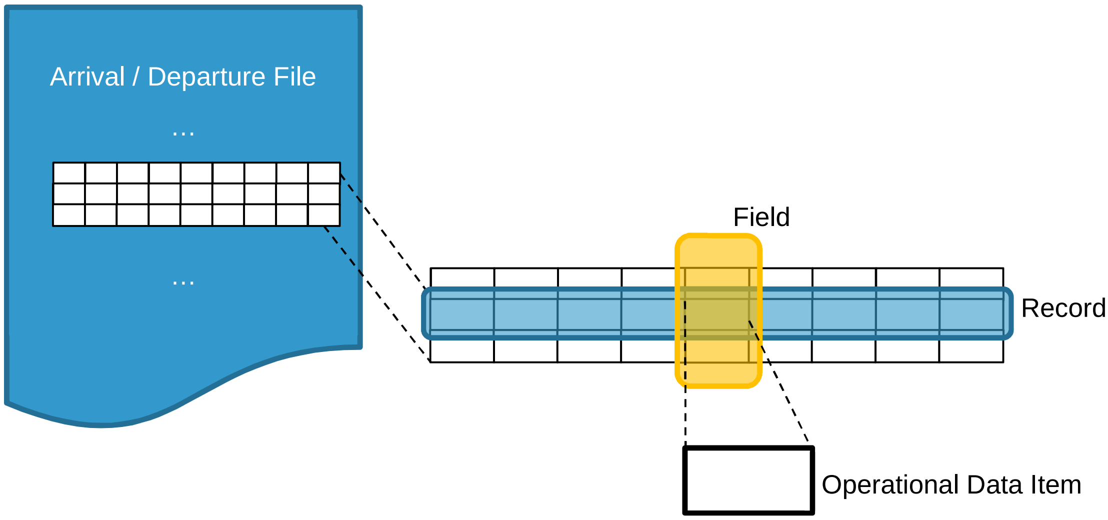

Figure 4 – Reporting File Nomenclature

The following requirements apply for the file format and syntax:

APDF-FFS-010-M Arrival and departure files **shall** be provided as separate .csv files with the following naming convention:

- APDF_<<ICAO-Location-Indicator>>_ARR_YYYYMM.csv
- APDF_<<ICAO-Location-Indicator>>_DEP_YYYYMM.csv

Example: APDF_LFBO_ARR_201402.csv.

APDF-FFS-020-M Arrival and departure files **shall** adhere to the specified file syntax summarised in Table 4. File syntax is understood as the combination of the operational data items that form a correctly structured set of records as defined in Section 3.3 of this specification.

APDF-FFS-030-M Records **shall** be separated by a CRLF (carriage return and line feed).

APDF-FFS-040-M Operational data items **shall** be separated by a semi-colon (;).

APDF-FFS-050-M Operational data items **shall** adhere to the following conventions:

- text entry qualifier: {none};
- number/numeric decimal symbol: decimal point (.);
- number/numeric 1000 separator: {none};

- timestamps
  - time standard: UTC or local time LT, both may be provided;
  - date delimiter: dash (-);
  - time delimiter: colon (:);
  - leading zeros: yes; and
  - general timestamp format: DD-MM-YYYY hh:mm:ss.

Table 4 shows the principal APDF file format structure. Requirements on each field/operational data item are listed throughout this specification.

APDF_ICAO_ARR_YYYYMM.csv

Field #	Field Header	ODI #
01	REG	ODI 01
02	ARCTYP	ODI 02
03	FLTID	ODI 03
04	ADEP_ICAO	
05	ADEP_IATA	ODI 06
06	ADES_ICAO	
07	ADES_IATA	
08	STA_UTC	ODI 10
09	STA_LT	
10	ALDT_UTC	ODI 13
11	ALDT_LT	
12	AIBT_UTC	ODI 14
13	AIBT_LT	
14	FLTRUL	ODI 04
15	FLTTYP	ODI 05
16	SIBT_UTC	ODI 22
17	SIBT_LT	
18	ARWY	ODI 07
19	ASTND	ODI 08
20	STATUS	ODI 18
21	RCNL	ODI 19
22	ATOC_UTC	ODI 20
23	ATOC_LT	
24	SVCTYP	ODI 23
25	IFPLID	ODI 24

APDF_ICAO_DEP_YYYYMM.csv

Field #	Field Header	ODI #
01	REG	ODI 01
02	ARCTYP	ODI 02
03	FLTID	ODI 03
04	ADEP_ICAO	
05	ADEP_IATA	ODI 06
06	ADES_ICAO	
07	ADES_IATA	
08	STD_UTC	ODI 09
09	STD_LT	
10	AOBT_UTC	ODI 11
11	AOBT_LT	
12	ATOT_UTC	ODI 12
13	ATOT_LT	
14	FLTRUL	ODI 04
15	FLTTYP	ODI 05
16	SOBT_UTC	ODI 21
17	SOBT_LT	
18	DRWY	ODI 07
19	DSTND	ODI 08
20	DLY1	
21	TIME1	
22	DLY2	
23	TIME2	
24	DLY3	ODI 15
25	TIME3	ODI 16
26	DLY4	
27	TIME4	
28	DLY5	
29	TIME5	
30	DE_ANTI_ICING	ODI 17
31	STATUS	ODI 18
32	RCNL	ODI 19
33	ATOC_UTC	ODI 20
34	ATOC_LT	
35	SVCTYP	ODI 23
36	IFPLID	ODI 24

Table 4 – APDF File Format and Syntax

3.3 Record Format and Syntax

For each file type (arrival or departure), the file consists of all operated flights and operational cancellations for the respective airport. Accordingly, three types of records exist:

- arrival record
- departure record; and
- (arrival or departure) operational cancellation record.

Dependent on the level of coordination (i.e. IATA Level 3, 2, or 1) of an airport, the record format (set of required ODIs) differs for an airport and the associated record type.

The composition of records is presented in Table 5 for arrival files and Table 6 for departure files. The field number represents the position in the record in accordance with Figure 4 and is a format/syntax requirement for the respective record.

Note:

- operational cancellations are reported as part of the arrival and departure file following the cancellation record format specified below.

APDF-RFS-010-M Records for arrival and departure files **shall** comply with the record format presented in Table 5 (arrival file record format) and Table 6 (departure file record format). In particular, the sequence of data fields in the record **shall** be maintained.

APDF-RFS-020-M Record field headers **shall** be compliant with the naming conventions throughout this specification (i.e. Table 4, Table 5, and Table 6).

Equally to ODIs, field headers **shall** be separated by a semi-colon (;) without any additional surrounding symbols or characters.

Note:

- the arrival files do not contain ODIs #15 and #17 (i.e. delay causes and durations), and ODI#17 (de/anti-icing information).
- the arrival and departure files make use of the same ODIs with the caveat that ODIs #01 to #08 have to be interpreted in the message context: e.g. ODI#07 refers to the runway used by the flight and hence must be distinguished as arrival runway (ARWY) and departure runway (DRWY) respectively.

APDF-RFS-030-M Operational data items allowing for the use of alternative representations, e.g. ICAO/IATA coding, UTC/local time **shall** be considered well formatted when at least one of the presentations is compliant with the specification..

For example, fields #4/5 (=ODI#06)): the departure airport ADEP is provided by either providing the ICAO location indicator (field #4), the IATA location indicator (field #5) or both.

3.3.1 Operational Cancellations (Arrival and Departure Record)

APDF-RFS-100-M a well-formatted cancellation record **shall** comprise all the required data items and – accordingly – all other items **shall not** be populated.

APDF-RFS-110-M level 3 airports **shall** provide all the required data items of a cancellation record (i.e. fields #2/ARCTYP, #3/FLTID, #4/5/ADEP, #6/7/ADES and

- for arrivals: fields #8/9/STA, #16/17/SIBT, #20/STATUS, #21/RCNL and #22/23/ATOC; or

- for departures: fields #8/9/STD, #16/17/SOBT, #31/STATUS, #32/RCNL and #33/34/ATOC.

APDF-RFS-120-R level 3 airports **should** provide the aircraft registration, field #1/REG, if available at the time of cancellation.

APDF-RFS-130-M level 2 and 1 airports **shall** provide the fields #2/ARCTYP, #3/FLTID, #4/5/ADEP, #6/7/ADES and

- for arrivals: fields #8/9/STA, #20/STATUS, #21/RCNL and #22/23/ATOC, or
- for departures: fields #8/9/STD, #20/STATUS, #21/RCNL and #22/23/ATOC

APDF-RFS-140-R level 2 and 1 airports **should** provide the following fields:

- aircraft registration, field #1/REG, if available at the time of cancellation;
- scheduled block times SIBT and SOBT, fields #16/17, if a facilitated schedule has been established.

APDF-RFS-150-HD all airports **should** provide as highly desirable fields:

- for arrivals: fields #24/SVCTYP and #25/IFPLID
- for departures: fields #35/SVCTYP and #36/IFPLID.

Note:

- If a non-standard IATA code for a given cancelled flight is provided or coordinated (e.g. by the air transport operator or ground handler) and the code cannot be cleared and/or converted into any of the standard IATA codes (i.e. AHM730, AHM731 or SSIM, c.f. Chapter 6), then the code 9999 shall be used in the field RCNL.
- If for a given cancelled flight neither the airline nor the ground handler provided any information regarding the reason for the cancellation, then the code ZZZZ shall be used in the field RCNL.
- If the actual time of cancellation is not available, the value 01-01-1990 00:00:00 shall be used in the field ATOC_UTC.

3.3.2 Arrival Record

APDF-RFS-200-M a well-formatted operated arrival flight record **shall** comprise all required data items and – accordingly – all other items **shall not** be populated.

Note: operational cancellations are specified above (i.e. 3.3.1).

APDF-RFS-210-M all airports **shall** provide fields #1/REG, #2/ARCTYP, #3/FLTID, #4/5/ADEP, #6/7/ADEP, #8/9/STA, #10/11/ALDT, #12/13/AIBT, #14/FLTRUL, #15/FLTTYP, #18/ARWY and #19/ASTND.

APDF-RFS-220-M level 3 airports **shall** provide field #16/17/SIBT only for flights under coordination procedures which received an airport slot.

APDF-RFS-230-R level 2 and 1 airports **should** provide field #16/17/SIBT, if a facilitated schedule has been established.

APDF-RFS-240-HD all airports **should** provide as highly desirable fields #24/SVCTYP and #25/IFPLID.

The following table depicts these requirements (i.e. mandatory and optional ODIs):

APDF_ICAO_ARR_YYYYMM.csv						
Field #	**Field Header**	**ODI #**	**L3 CAN**	**L3 ARR**	**L21 CAN**	**L21 ARR**
1	REG	01	R	M	R	M
2	ARCTYP	02	M	M	M	M
3	FLTID	03	M	M	M	M
4	ADEP_ICAO	06	M	M	M	M
5	ADEP_IATA					
6	ADES_ICAO					
7	ADES_IATA					
8	STA_UTC	10	M	M	M	M
9	STA_LT					
10	ALDT_UTC	13	--	M	--	M
11	ALDT_LT					
12	AIBT_UTC	14	--	M	--	M
13	AIBT_LT					
14	FLTRUL	04	--	M	--	M
15	FLTTYP	05	--	M	--	M
16	SIBT_UTC	22	M	M	R	R
17	SIBT_LT					
18	ARWY	07	--	M	--	M
19	ASTND	08	--	M	--	M
20	STATUS	18	M	--	M	--
21	RCNL	19	M	--	M	--
22	ATOC_UTC	20	M	--	M	--
23	ATOC_LT					
24	SVCTYP	23	HD	HD	HD	HD
25	IFPLID	24	HD	HD	HD	HD

Table 5 – Arrival File Record Format

CAN – cancelled arrival record; ARR – operated arrival flight record
IATA level of coordination: L3: Level 3 airport; L21: Level 2 or 1 airport

--	not allowed
M	mandatory
R	recommended
HD	highly desirable

3.3.3 Departure Record

APDF-RFS-300-M a well-formatted operated departure flight record **shall** comprise all required data items and – accordingly – all other items **shall not** be populated.

Note: operational cancellations are specified above (i.e. 3.3.1).

APDF-RFS-310-M all airports **shall** provide fields #1/REG, #2/ARCTYP, #3/FLTID, #4/5/ADEP, #6/7/ADEP, #8/9/STD, #10/11/AOBT, #12/13/ATOT, #14/FLTRUL, #15/FLTTYP, #18/DRWY, #19/DSTND, and #30/DE_ANT_ICING.

APDF-RFS-320-M level 3 airports **shall** provide field #16/17/SOBT only for flights under coordination procedures which received an airport slot.

APDF-RFS-330-R level 2 and 1 airports **should** provide field #16/17/SOBT, if a facilitated schedule has been established.

APDF-RFS-340-M operated departure flights **shall** be considered <u>delayed</u> when the delay threshold is breached, i.e. AOBT ≥ STD + 00:04:00 (240 seconds). If the delay condition holds, operated flight records **shall** additionally comprise <u>at least</u> one delay cause (i.e. field #20/DLY1) and the associated delay duration (i.e. field #21/TIME1).

APDF-RFS-350-R to allow for a proper accounting of various delay contributions, other delay causes and their duration (fields #22-29) **should** be provided ensuring that the reported delay contributions (i.e. TIME1+TIME2+TIME3+TIME4+TIME5, as applicable) does not exceed the actual delay (i.e. AOBT – STD).

Note:

- If a non-standard IATA code or sub-code for a given delayed flight is provided or coordinated (e.g. by the air transport operator or ground handler) and the code cannot be cleared and/or converted into any of the standard IATA codes (i.e. AHM730 or AHM731, c.f. Chapter 6), then the code 999 shall be used.
- If for a given delayed flight neither the air transport operator nor the ground handler (or any other appropriate entity) provided information regarding the cause of the delay, then the code ZZZ shall be used in field DLY1. The associated field TIME1 shall be populated with the result of the actual delay, in minutes (AOBT minus STD).

APDF-RFS-360-HD all airports **should** provide as highly desirable fields #35/SVCTYP and #36/IFPLID.

The following table depicts these requirements (i.e. mandatory and optional ODIs):

APDF_ICAO_DEP_YYYYMM.csv						
Field #	**Field Header**	**ODI #**	**L3 CAN**	**L3 DEP**	**L21 CAN**	**L21 DEP**
1	REG	01	R	M	R	M
2	ARCTYP	02	M	M	M	M
3	FLTID	03	M	M	M	M
4	ADEP_ICAO	06	M	M	M	M
5	ADEP_IATA					
6	ADES_ICAO					
7	ADES_IATA					
8	STD_UTC	09	M	M	M	M
9	STD_LT					
10	AOBT_UTC	11	--	M	--	M
11	AOBT_LT					
12	ATOT_UTC	12	--	M	--	M
13	ATOT_LT					
14	FLTRUL	04	--	M	--	M
15	FLTTYP	05	--	M	--	M
16	SOBT_UTC	21	M	M	R	R
17	SOBT_LT					
18	DRWY	07	--	M	--	M
19	DSTND	08	--	M	--	M
20	DLY1	15 16	--	M/delay	--	M/delay
21	TIME1					
22	DLY2					
23	TIME2					
24	DLY3					
25	TIME3					
26	DLY4					
27	TIME4					
28	DLY5					
29	TIME5					
30	DE_ANTI_ICING	17	--	M	--	M
31	STATUS	18	M	--	M	--
32	RCNL	19	M	--	M	--
33	ATOC_UTC	20	M	--	M	--
34	ATOC_LT					
35	SVCTYP	23	HD	HD	HD	HD
36	IFPLID	24	HD	HD	HD	HD

Table 6 – Departure File Record Format

CAN – cancelled departure record; DEP – operated departure flight record
IATA level of coordination: L3: Level 3 airport; L21: Level 2 or 1 airport

--	not allowed
M	mandatory
R	recommended
HD	highly desirable

4. Additional Quality Assurance Measure

Chapter 2 provides an overview of the airport operator data flow and its data processing steps. This specification has been drawn up to establish quality assurance principles ensuring a consistent approach to process and data quality of the APDF. The validation activities surrounding the APDF and the calculation of certain performance indicators (e.g. additional ASMA time) require the association of the data submitted by the reporting entity with data derived from other sources.

To facilitate unambiguous association of each record received the following set of fields play a major role and are subject to further quality controls by validating permissible value domains or other quality criteria (i.e. relevance, consistency and coherence) with supporting secondary data sources.

The key association fields include:

- ODI#01 – REG, aircraft registration
- ODI#02 – ARCTYP, aircraft type
- ODI#03 – FLTID, flight identifier
- ODI#06 – ADEP/ ADES, encoded airport of departure and destination
- ODI#11 – AOBT Actual Off-Block Time
- ODI#12 – ATOT, Actual Take Off Time
- ODI#13 – ALDT, Actual Landing Time
- ODI#14 – AIBT, Actual In-Block Time

For the purpose of enriching the level of association across the different sources, the IFPS flight plan identifier, ODI#24 – IFPLID, provides a common denominator and identification key. In that respect, ODI#24 – IFPLID has been identified as a highly desirable data item. This ODI will be used to validate the flight identification with the flight plan held by the IFPS and the aforementioned set of fields.

5. Operational Data Item Specifications

APDF-ODI-000-M The operational data item specification presented in this Chapter **shall** be applicable for all data records provided under this specification.

Note:

- The requirements for providing the operational data item as part of the respective arrival, departure or cancellation record are specified in Section 3.3.
- While the provision of the operational data items ODI 23, IATA Service Type, and ODI 24, IFPS Flight Plan ID, are highly desirable in accordance with Section 3.3, this specification sets-out mandatory data formatting and quality checks as specified in this chapter,

5.1 ODI 01 Aircraft Registration (APDF-ODI-001-M)

ODI#	01
ID	REG
Version ID / Version date	00-11 / 04-09-2014
Item	aircraft registration
Acronym(s) [Field Header(s)]	REG
IR390/13 Annex V ODI	3.2.2.a
IR390/13 Annex V Definition	3.1.d 'aircraft registration' means the alphanumerical characters corresponding to the actual registration of the aircraft
Alternative definition	The nationality or common mark and registration mark of the aircraft, not exceeding 7 alphanumeric characters and without hyphens or symbols.
Source	ICAO Doc 4444 - Air Traffic Management (PANS-ATM)
Background info	ICAO Annex 7 - Aircraft Nationality and Registration Marks
Scope	arrivals and departures; recommended for operational cancellations.
Data format	
Data type	alphanumeric string
Data note	not exceeding 7 alphanumeric characters and without hyphens or symbols or any other special character.
Permissible value domain	[A-Z0-9]{1,7}
Example	ECJON
Quality Checks	
Relevance	key association field key item for duplicated flights check[1]
Accuracy	aircraft type and registration shall be consistent with PRISME Fleet[2]
Coherence and Comparability	≥98% of completeness for IFR flights (flight rules I, Y or Z)
Other Checks	
Notes	

[1] duplicated flights: flight records for which association with other data sources fails (e.g. NM database) and which show identical airport of departure/destination (ADEP/ADES), aircraft registration (REG), and time stamps (STA/STD, ALDT/ATOT, AIBT/AOBT), excluding operational cancellations.
[2] https://www.eurocontrol.int/services/prisme-fleet

5.2 ODI 02 Aircraft Type (APDF-ODI-002-M)

ODI#	02
ID	ARCTYP
Version ID / Version date	00-11 / 04-09-2014
Item	Aircraft type
Acronym(s) [Field Header(s)]	ARCTYP
IR390/13 Annex V ODI	3.2.2.b and 3.2.3.b
IR390/13 Annex V Definition	3.1.e 'aircraft type' means an aircraft type designator (up to four characters) as indicated in ICAO Doc 8643
Alternative definition	
Source	
Background info	ICAO Doc 8643 - Aircraft Type Designators[1]
Scope	arrivals, departures, and operational cancellations.
Data format	
Data type	alphanumeric string
Data note	not exceeding 4 alphanumerical characters and without hyphens or symbols or any other special character.
Permissible value domain	[A-Z0-9]{1,4}
Example	B738
Quality Checks	
Relevance	key association field
Accuracy	aircraft type and registration shall be consistent with PRISME Fleet[2]
Coherence and Comparability	≥ 98% of completeness for IFR flights (flight rules I, Y or Z) and operational cancellations
Other Checks	
Notes	

[1] http://www.icao.int/publications/DOC8643
[2] https://www.eurocontrol.int/services/prisme-fleet

5.3 ODI 03 Flight Identifier (APDF-ODI-003-M)

ODI#	03
ID	FLTID
Version ID / Version date	00-11 / 04-09-2014
Item	Flight identifier
Acronym(s) [Field Header(s)]	FLTID
IR390/13 Annex V ODI	3.2.2.c and 3.2.3.a
IR390/13 Annex V Definition	3.1.f 'flight identifier' means a group of alphanumeric characters used to identify a flight. Item 7 of the ICAO flight plan.
Alternative definition	Aircraft Identification (ARCID): May be the registration marking of the aircraft, or the ICAO designator of the aircraft operator followed by the flight identification.
Source	EUROCONTROL Specification for ATS Data Exchange Presentation (ADEXP)
Background info	ICAO Doc 4444 - Air Traffic Management (PANS-ATM)
Scope	arrivals, departures, and operational cancellations
Data format	
Data type	alphanumeric string
Data note	no space between the operator's designator and the aircraft registration or flight identification.
Permissible value domain	[A-Z0-9]{1,7}
Example	IBE360K or ECJON
Quality Checks	
Relevance	key association field
Accuracy	
Coherence and Comparability	≥ 98% of completeness for IFR flights (flight rules I, Y or Z) and operational cancellations
Other Checks	
Notes	

5.4 ODI 04 Flight rules (APDF-ODI-004-M)

ODI#	04
ID	FLTRUL
Version ID / Version date	00-11 / 04-09-2014
Item	Flight rules
Acronym(s) [Field Header(s)]	FLTRUL
IR390/13 Annex V ODI	3.2.2.h
IR390/13 Annex V Definition	3.1.o 'flight rules' means the rules used in conducting the flight. 'IFR' for aircraft flying according to instrument flight rules, as defined in Annex 2 of the Chicago Convention or 'VFR' for aircraft flying according to visual flight rules as defined in the same Annex. 'Operational Air Traffic (OAT)' refers to State aircraft not following the rules defined in Annex 2 of the Chicago Convention. Item 8 of the ICAO flight plan.
Alternative definition	
Source	
Background info	ICAO Doc 4444 - Air Traffic Management (PANS-ATM)
Scope	arrivals and departures
Data format	
Data type	alphanumeric string
Data note	letters to denote the category of flight rules as defined in Appendix 2 of ICAO Doc 4444: I — entire flight operated under the IFR V — entire flight operated under the VFR Y — flight initially operated under the IFR, followed by one or more subsequent changes of flight rules Z — flight initially operated under the VFR, followed by one or more subsequent changes of flight rules
Permissible value domain	[IVYZ]{1}
Example	I
Quality Checks	
Relevance	key item for quality checks
Accuracy	
Coherence and Comparability	≥ 98% of completeness for operated flights
Other Checks	
Notes	

5.5 ODI 05 Flight type (APDF-ODI-005-M)

ODI#	05
ID	FLTTYP
Version ID / Version date	00-11 / 04-09-2014
Item	Flight type
Acronym(s) [Field Header(s)]	FLTTYP
IR390/13 Annex V ODI	3.2.2.h
IR390/13 Annex V Definition	3.1.p 'flight type' means the type of flight as defined in Appendix 2 of ICAO Doc 4444 (15th Edition — June 2007)
Alternative definition	
Source	
Background info	ICAO Doc 4444 - Air Traffic Management (PANS-ATM)
Scope	arrivals and departures
Data format	
Data type	alphanumeric string
Data note	Letters to denote the type of flight as defined in Appendix 2 of ICAO Doc 4444: S — scheduled air service N — non-scheduled air transport operation G — general aviation M — military X — other than any of the defined categories above
Permissible value domain	[SNGMX]{1}
Example	S
Quality Checks	
Relevance	key item for quality checks
Accuracy	
Coherence and Comparability	≥98% of completeness for IFR flights (flight rules I, Y or Z)
Other Checks	
Notes	

5.6 ODI 06 Encoded airport of departure and of destination (APDF-ODI-006-M)

ODI#	06
ID	AD
Version ID / Version date	00-11 / 04-09-2014
Item	Encoded aerodrome of departure/destination
Acronym(s) [Field Header(s)]	ADEP / ADES
IR390/13 Annex V ODI	3.2.2.d and 3.2.3.e
IR390/13 Annex V Definition	3.1.g 'encoded aerodrome of departure/destination' mean the code of the airport using the ICAO four-letter or the IATA three-letter airport designator
Alternative definition	
Source	
Background info	ICAO Doc 7910 - Location Indicators
Scope	arrivals, departures, and operational cancellations
Data format	
Data type	alphanumeric string
Data note	either ICAO (preferred) or IATA (alternative) code is sufficient but both may be provided.
Permissible value domain	[A-Z]{4} or [A-Z]{3}
Example	LEMD or MAD
Quality Checks	
Relevance	key association field
Accuracy	Location indicator codes shall be consistent with ICAO Doc 7910 - Location Indicators.
Coherence and Comparability	≥ 98% of completeness for IFR flights (flight rules I, Y or Z) and operational cancellations
Other Checks	
Notes	

5.7 ODI 07 Arrival and departure runway designator (APDF-ODI-007-M)

ODI#	07
ID	RWY
Version ID / Version date	00-11 / 04-09-2014
Item	Runway designator
Acronym(s) [Field Header(s)]	DRWY / ARWY
IR390/13 Annex V ODI	3.2.2.j
IR390/13 Annex V Definition	3.1.r ‘arrival runway designator’ and ‘departure runway designator’ mean the ICAO designator of the runway used for landing and for take-off (e.g. 10L);
Alternative definition	
Source	
Background info	ICAO Annex 14 Volume I - Aerodrome Design and Operations
Scope	arrivals and departures
Data format	
Data type	alphanumeric string
Data note	a runway designator consists of a two-digit number and on parallel runways shall be supplemented with a letter. On a single runway, dual parallel runways and triple parallel runways the two-digit number shall be the whole number nearest the one-tenth of the magnetic North when viewed from the direction of approach. On four or more parallel runways, one set of adjacent runways shall be numbered to the nearest one-tenth magnetic azimuth and the other set of adjacent runways numbered to the next nearest one-tenth of the magnetic azimuth. When the above rule would give a single digit number, it shall be preceded by a zero. In the case of parallel runways, each runway designation number shall be supplemented by a letter as follows, in the order shown from left to right when viewed from the direction of approach: — two parallel runways: “L” “R”; — three parallel runways: “L” “C” “R”; — four parallel runways: “L” “R” “L” “R”; — five parallel runways: “L” “C” “R” “L” “R” or “L” “R” “L” “C” “R”; and — six parallel runways: “L” “C” “R” “L” “C” “R”.
Permissible value domain	[0-9]{2}[LRC]{0,1} or [H]{1}[[A-Z0-9]{0,2}
Example	08R or H
Quality Checks	
Relevance	key item for (K)PIs calculation
Accuracy	runway designators shall be consistent with AIP AD2
Coherence and Comparability	≥ 98% of completeness for IFR flights (flight rules I, Y or Z)
Other Checks	
Notes	

5.8 ODI 08 Arrival and departure stand (APDF-ODI-008-M)

ODI#	08
ID	STND
Version ID / Version date	00-11 / 04-09-2014
Item	Stand
Acronym(s) [Field Header(s)]	ASTND / DSTND
IR390/13 Annex V ODI	3.2.2.k
IR390/13 Annex V Definition	3.1.3.s ‘arrival stand’ means the designator of the first parking position where the aircraft was parked upon arrival 3.1.3.t ‘departure stand’ means the designator of the last parking position where the aircraft was parked before departing from the airport
Alternative definition	
Source	
Background info	
Scope	arrivals and departures
Data format	
Data type	alphanumeric string
Data note	
Permissible value domain	{1,32}
Example	A69 or TG/LA (c.f. notes below)
Quality Checks	
Relevance	key item for (K)PIs calculation
Accuracy	stand (parking position) designators shall be consistent with AIP AD2
Coherence and Comparability	≥ 98% of completeness for IFR flights (flight rules I, Y or Z)
Other Checks	
Notes	For touch-and-go and low approach manoeuvres, the unique code **TG/LA** shall be provided in fields ASTND and DSTND.

5.9 ODI 09 Schedule Time of Departure (off-block) (APDF-ODI-009-M)

ODI#	09
ID	STD
Version ID / Version date	00-11 / 04-09-2014
Item	Scheduled time of departure (off-block)
Acronym(s) [Field Header(s)]	STD_UTC / STD_LT
IR390/13 Annex V ODI	3.2.2.e and 3.2.3.c
IR390/13 Annex V Definition	3.1.i 'scheduled time of departure (off-block)' means date and time when a flight is scheduled to depart from the departure stand
Alternative definition	
Source	
Background info	
Scope	departures and departure operational cancellations
Data format	
Data type	DateTime DD-MM-YYYY hh:nn:ss
Data note	time in UTC or LT (Local Time) is sufficient but both may be provided.
Permissible value domain	
Example	27-05-2014 09:05:00
Quality Checks	
Relevance	key item for (K)PIs calculation key item for quality checks key item for duplicated flights check[1]
Accuracy	
Coherence and Comparability	≥ 98% of completeness for IFR flights (flight rules I, Y or Z) and operational cancellations ≥ 95% of associated flights (Airport vs Airline) with values in tolerance interval]-1,+1[min ≥ 99% of associated flights (Airport vs Airline) with values in tolerance interval [-120,+120] min
Other Checks	
Notes	

[1] duplicated flights: flight records for which association with other data sources fails (e.g. NM database) and which show identical airport of departure/destination (ADEP/ADES), aircraft registration (REG), and time stamps (STA/STD, ALDT/ATOT, AIBT/AOBT), excluding operational cancellations.

5.10 ODI 10 Schedule Time of Arrival (in-block) (APDF-ODI-010-M)

ODI#	10
ID	STA
Version ID / Version date	00-11 / 04-09-2014
Item	Scheduled time of arrival (on-block)
Acronym(s) [Field Header(s)]	STA_UTC / STA_LT
IR390/13 Annex V ODI	3.2.2.f and 3.2.3.d
IR390/13 Annex V Definition	3.1.m 'scheduled time of arrival (on-block)' means the date and time when a flight is scheduled to arrive at the arrival stand
Alternative definition	
Source	
Background info	
Scope	arrivals and arrival operational cancellations
Data format	
Data type	DateTime DD-MM-YYYY hh:nn:ss
Data note	time in UTC or LT (Local Time) is sufficient but both may be provided.
Permissible value domain	
Example	15-09-2013 08:05:00
Quality Checks	
Relevance	key item for quality checks key item for duplicated flights check[1]
Accuracy	
Coherence and Comparability	≥ 98% of completeness for IFR flights (flight rules I, Y or Z) and operational cancellations ≥ 95% of associated flights (Airport vs Airline) with values in tolerance interval]-1,+1[min ≥ 99% of associated flights (Airport vs Airline) with values in tolerance interval [-120,+120] min
Other Checks	
Notes	

[1] duplicated flights: flight records for which association with other data sources fails (e.g. NM database) and which show identical airport of departure/destination (ADEP/ADES), aircraft registration (REG), and time stamps (STA/STD, ALDT/ATOT, AIBT/AOBT), excluding operational cancellations.

5.11 ODI 11 Actual Off-Block Time (APDF-ODI-011-M)

ODI #	11
ID	AOBT
Version ID / Version date	00-11 / 04-09-2014
Item	Actual off-block time
Acronym(s) [Field Header(s)]	AOBT_UTC / AOBT_LT
IR390/13 Annex V OD	3.2.2.g
IR390/13 Annex V Definition	3.1.h.i 'actual off-block time' means the actual date and time the aircraft has vacated the parking position (pushed back or on its own power)
Alternative definition	Time the aircraft pushes back / vacates the parking position. (Equivalent to Airline / Handlers ATD – Actual Time of Departure, ACARS=OUT)
Source	EUROCONTROL, ACI, IATA - The Manual. Airport CDM Implementation.
Background info	
Scope	departures
Data format	
Data type	DateTime DD-MM-YYYY hh:nn:ss
Data note	time in UTC or LT (Local Time) is sufficient but both may be provided.
Permissible value domain	
Example	06-03-2014 09:05:00
Quality Checks	
Relevance	key association field key item for (K)PIs calculation key item for duplicated flights check[1] key item for chronological order check
Accuracy	100% flights with AOBT≤ ATOT ≥ 95% flights with AXOT[2] > 0 min ≥ 95% flights with AXOT < 60min
Coherence and Comparability	≥ 98% of completeness for IFR flights (flight rules I, Y or Z) ≥ 95% of associated flights with (Airport vs Airline) values in tolerance interval [-3,+3] min ≥ 99% of associated flights with (Airport vs Airline) values in tolerance interval [-120,+120] min
Other Checks	
Notes	For both, touch-and-go and low approach manoeuvres, AOBT shall be equal to ATOT.

[1] duplicated flights: flight records for which association with other data sources fails (e.g. NM database) and which show identical airport of departure/destination (ADEP/ADES), aircraft registration (REG), and time stamps (STA/STD, ALDT/ATOT, AIBT/AOBT), excluding operational cancellations.
[2] AXOT: Actual Taxi-Out Time. Metric ATOT – AOBT

5.12 ODI 12 Actual Take Off Time (APDF-ODI-012-M)

ODI#	12
ID	ATOT
Version ID / Version date	00-11 / 04-09-2014
Item	Actual take-off time
Acronym(s) [Field Header(s)]	ATOT_UTC / ATOT_LT
IR390/13 Annex V ODI	3.2.2.g
IR390/13 Annex V Definition	3.1.k 'actual take off time' means the date and time that an aircraft has taken off from the runway (wheels-up)
Alternative definition	The time that an aircraft takes off from the runway. (Equivalent to ATC ATD – Actual Time of Departure, ACARS = OFF)
Source	EUROCONTROL, ACI, IATA - The Manual. Airport CDM Implementation.
Background info	
Scope	departures
Data format	
Data type	DateTime DD-MM-YYYY hh:nn:ss
Data note	time in UTC or LT (Local Time) is sufficient but both may be provided.
Permissible value domain	
Example	11-01-2008 09:10:00
Quality Checks	
Relevance	key item for (K)PIs calculation key association field key item for duplicated flights check[1] key item for chronological order check
Accuracy	100% flights with AOBT≤ ATOT ≥ 95% flights with AXOT[2] > 0 min ≥ 95% flights with AXOT < 60min
Coherence and Comparability	≥ 98% of completeness for IFR flights (flight rules I, Y or Z) ≥ 95% of associated flights with (Airport vs Airline) values in tolerance interval [-3,+3] min ≥ 95% of associated flights with (Airport vs NM) values in tolerance interval [-3,+3] min ≥ 99% of associated flights with (Airport vs Airline) values in tolerance interval [-120,+120] min ≥ 99% of associated flights with (Airport vs NM) values in tolerance interval [-120,+120] min
Other Checks	
Notes	

[1] duplicated flights: flight records for which association with other data sources fails (e.g. NM database) and which show identical airport of departure/destination (ADEP/ADES), aircraft registration (REG), and time stamps (STA/STD, ALDT/ATOT, AIBT/AOBT), excluding operational cancellations.
[2] AXOT: Actual Taxi-Out Time. Metric ATOT – AOBT

5.13 ODI 13 Actual Landing Time (APDF-ODI-013-M)

ODI#	13
ID	ALDT
Version ID / Version date	00-11 / 04-09-2014
Item	Actual landing time
Acronym(s) [Field Header(s)]	ALDT_UTC / ALDT_LT
IR390/13 Annex V ODI	3.2.2.g
IR390/13 Annex V Definition	3.1.l 'actual landing time' means the actual date and time when the aircraft has landed (touch down)
Alternative definition	The time that an aircraft lands on a runway. (Equivalent to ATC ATA – Actual Time of Arrival, ACARS = ON)
Source	EUROCONTROL, ACI, IATA - The Manual. Airport CDM Implementation.
Background info	
Scope	arrivals
Data format	
Data type	DateTime DD-MM-YYYY hh:nn:ss
Data note	time in UTC or LT (Local Time) is sufficient but both may be provided.
Permissible value domain	
Example	11-01-2008 09:05:00
Quality Checks	
Relevance	key item for (K)PIs calculation key association field key item for duplicated flights check[1] key item for chronological order check
Accuracy	100% flights with ALDT≤ AIBT ≥ 95% flights with AXIT [2] > 0 min ≥ 95% flights with AXIT < 60min
Coherence and Comparability	≥ 98% of completeness for IFR flights (flight rules I, Y or Z) ≥ 95% of associated flights with (Airport vs Airline) values in tolerance interval [-3,+3] min ≥ 95% of associated flights with (Airport vs NM) values in tolerance interval [-3,+3] min ≥ 99% of associated flights with (Airport vs Airline) values in tolerance interval [-120,+120] min ≥ 99% of associated flights with (Airport vs NM) values in tolerance interval [-120,+120] min
Other Checks	
Notes	

[1] duplicated flights: flight records for which association with other data sources fails (e.g. NM database) and which show identical airport of departure/destination (ADEP/ADES), aircraft registration (REG), and time stamps (STA/STD, ALDT/ATOT, AIBT/AOBT), excluding operational cancellations.
[2] AXIT: Actual Taxi-In Time. Metric AIBT – ALDT

5.14 ODI 14 Actual In-Block Time (APDF-ODI-014-M)

ODI#	14
ID	AIBT
Version ID / Version date	00-11 / 04-09-2014
Item	Actual on-block time
Acronym(s) [Field Header(s)]	AIBT_UTC / AIBT_LT
IR390/13 Annex V ODI	3.2.2.g
IR390/13 Annex V Definition	3.1.n 'actual on-block time' means the date and time when the parking brakes have been engaged at the arrival stand
Alternative definition	The time that an aircraft arrives in-blocks. (Equivalent to Airline/Handler ATA – Actual Time of Arrival, ACARS = IN)
Source	EUROCONTROL, ACI, IATA - The Manual. Airport CDM Implementation.
Background info	
Scope	arrivals
Data format	
Data type	DateTime DD-MM-YYYY hh:nn:ss
Data note	time in UTC or LT (Local Time) is sufficient but both may be provided.
Permissible value domain	
Example	11-01-2008 09:10:00
Quality Checks	
Relevance	key association field key item for duplicated flights check [1] key item for chronological order check
Accuracy	100% flights with ALDT≤ AIBT ≥ 95% flights with AXIT > 0 min ≥ 95% flights with AXIT < 60min
Coherence and Comparability	≥ 98% of completeness for IFR flights (flight rules I, Y or Z) ≥ 95% of associated flights with (Airport vs Airline) values in tolerance interval [-3,+3] min ≥ 99% of associated flights with (Airport vs Airline) values in tolerance interval [-120,+120] min
Other Checks	
Notes	For both, touch-and-go and low approach manoeuvres, AIBT shall be equal to ALDT.

[1] duplicated flights: flight records for which association with other data sources fails (e.g. NM database) and which show identical airport of departure/destination (ADEP/ADES), aircraft registration (REG), and time stamps (STA/STD, ALDT/ATOT, AIBT/AOBT), excluding operational cancellations.

5.15 ODI 15 Delay causes (APDF-ODI-015-M)

ODI#	15
ID	RDLY
Version ID / Version date	00-11 / 04-09-2014
Item	Delay cause
Acronym(s) [Field Header(s)]	DLY1, DLY2, DLY3, DLY4, DLY5
IR390/13 Annex V ODI	3.2.2.l
IR390/13 Annex V Definition	3.1.u ‘delay causes’ means the standard IATA delay codes as defined in Section F of CODA Digest Annual 2011 ‘Delays to Air Transport in Europe’ with the duration of the delay. Where several causes may be attributable to flight delays, a list of those causes shall be provided
Alternative definition	
Source	
Background info	IATA Airport Handling Manual (AHM730 and AHM731)
Scope	departures where the delay condition holds (AOBT>STD)
Data format	
Data type	alphanumeric string
Data note	delay codes can be provided in the following code formats: — 2-number code and 1-letter sub-code form IATA's Airport Handling Manual 730 and 731 (preferred) — 2-letter code and 1-letter sub-code form IATA's Airport Handling Manual 730 and 731 (alternative)
Permissible value domain	[0-9]{2} or [A-Z]{2} or [0-9A-Z]{3} or [A-Z]{3}
Example	89M
Quality Checks	
Relevance	key item for (K)PIs calculation
Accuracy	IATA AHM730 codes and AHM731 sub-codes check
Coherence and Comparability	100% completeness of TIMEn if DLYn is not null (for n = 1, 2, 3, 4, and 5) % Completeness check for flights on NM database where AOBT≥STD+00:04:00 and FLTTYP=[S] % Completeness check for flights on NM database where AOBT≥STD+00:04:00 and FLTTYP=[N]
Other Checks	
Notes	— If delay code clearing is not possible, the special code 999 shall be used as delay cause code. — If no delay code information is available/attainable, the special code ZZZ shall be provided.

5.16 ODI 16 Duration of the delay (APDF-ODI-016-M)

ODI#	16
ID	TIME
Version ID / Version date	00-11 / 04-09-2014
Item	Duration of the delay
Acronym(s) [Field Header(s)]	TIME1, TIME2, TIME3, TIME4, TIME5
IR390/13 Annex V ODI	3.2.2.l
IR390/13 Annex V Definition	3.1.u 'delay causes' means the standard IATA delay codes as defined in Section F of CODA Digest Annual 2011 'Delays to Air Transport in Europe' with the duration of the delay. Where several causes may be attributable to flight delays, a list of those causes shall be provided
Alternative definition	
Source	
Background info	IATA Airport Handling Manual (AHM730 and AHM731)
Scope	departures where the delay condition holds (AOBT>STD)
Data format	
Data type	integer
Data note	minutes of encountered delay
Permissible value domain	[0-9]{1,4}
Example	162
Quality Checks	
Relevance	key item for (K)PIs calculation
Accuracy	(AOBT - STD) ≥ (TIME1+TIME2+TIME3+TIME4+TIME5)
Coherence and Comparability	100% completeness of TIMEn if DLYn is not null (n = 1, 2, 3, 4, 5) % Completeness check for flights where AOBT≥STD+00:04:00 and FLTTYP=[S] % Completeness check for flights where AOBT≥STD+00:04:00 and FLTTYP=[N]
Other Checks	
Notes	

5.17 ODI 17 De-icing and anti-icing information (APDF-ODI-017-M)

ODI#	17
ID	DE-ANTI-ICING
Version ID / Version date	00-11 / 04-09-2014
Item	De-icing or anti-icing information
Acronym(s) [Field Header(s)]	DE_ANTI_ICING
IR390/13 Annex V ODI	3.2.2.m
IR390/13 Annex V Definition	3.1.v 'de-icing or anti-icing information' means indication if de-icing or anti-icing operations occurred and if yes, where (before leaving the departure stand or in a remote position after departing the stand, i.e. after off-block)
Alternative definition	
Source	
Background info	
Scope	departures
Data format	
Data type	alphanumeric string
Data note	A — after AOBT B — before AOBT N — no de/anti-icing process Z — no de/anti-icing information available
Permissible value domain	[ABNZ]{1}
Example	A
Quality Checks	
Relevance	key item for (K)PIs calculation
Accuracy	
Coherence and Comparability	100% of completeness for operated flights
Other Checks	
Notes	If no de/anti-icing information is available, the value Z shall be used.

5.18 ODI 18 Operational cancellation information (APDF-ODI-018-M)

ODI#	18
ID	STATUS
Version ID / Version date	00-11 / 04-09-2014
Item	Cancellation status
Acronym(s) [Field Header(s)]	STATUS
IR390/13 Annex V ODI	
IR390/13 Annex V Definition	3.1.w 'Operational cancellation' means an arrival or departure scheduled flight to which the following conditions apply: — the flight received an airport slot, and — the flight was confirmed by the air carrier the day before operations and/or it was contained in the daily list of flight schedules produced by the airport operator the day before of operations, but — the actual landing or take-off never occurred.
Alternative definition	
Source	
Background info	
Scope	operational cancellations
Data format	
Data type	alphanumeric string
Data note	C—Cancelled [Blank] — not Cancelled
Permissible value domain	[C]{1}
Example	C
Quality Checks	
Relevance	
Accuracy	
Coherence and Comparability	number of operational cancellations where (SOBT-12h) ≤ ATOC < (SOBT-3h) number of operational cancellations where (SOBT-3h) ≤ ATOC < SOBT number of operational cancellations where ATOC ≥ SOBT
Other Checks	
Notes	

5.19 ODI 19 Reason for Cancellation (APDF-ODI-019-M)

ODI#	18
ID	RCNL
Version ID / Version date	00-11 / 04-09-2014
Item	Reason for cancellation
Acronym(s) [Field Header(s)]	RCNL
IR390/13 Annex V ODI	3.2.3.g
IR390/13 Annex V Definition	
Alternative definition	
Source	
Background info	
Scope	operational cancellations
Data format	
Data type	alphanumeric string
Data note	cancellation reason codes can be provided in the following code formats: — 2-number code and 1-letter sub-code form IATA's Airport Handling Manual 730 and 731 (preferred) — 2-letter code and 1-letter sub-code form IATA's Airport Handling Manual 730 and 731 (alternative) — 4-letter code from IATA's Standard Schedules Information Manual
Permissible value domain	[0-9]{2} or [A-Z]{2} or [0-9A-Z]{3} or [A-Z]{3} or [A-Z]{4} or [9]{4}
Example	TECH
Quality Checks	
Relevance	
Accuracy	IATA AHM730, AHM731 and SSIM codes check
Coherence and Comparability	100% completeness for operational cancellations
Other Checks	
Notes	— If reason clearing is not possible, the special code 9999 shall be used as encoded reason for cancellation. — If no reason of cancellation is available/attainable, the special code ZZZZ shall be provided.

5.20 ODI 20 Actual Time Of Cancellation (APDF-ODI-020-M)

ODI#	20
ID	ATOC
Version ID / Version date	00-11 / 04-09-2014
Item	Actual time of cancellation
Acronym(s) [Field Header(s)]	ATOC_UTC / ATOC_LT
IR390/13 Annex V ODI	3.2.3.h
IR390/13 Annex V Definition	3.1.x 'actual time of cancellation' means the actual date and time when an arrival or departure of a scheduled flight was cancelled
Alternative definition	Date and time when the flight is not considered anymore in the planned demand of the airport system (AODB[1] and/or RMS[2]).
Source	
Background info	
Scope	operational cancellations
Data format	
Data type	DateTime DD-MM-YYYY hh:nn:ss
Data note	time in UTC or LT (Local Time) is sufficient but both may be provided.
Permissible value domain	
Example	11-11-2014 09:05:00
Quality Checks	
Relevance	
Accuracy	
Coherence and Comparability	100% completeness for operational cancellations number of operational cancellations where (STD-12h) ≤ ATOC < (STD-3h) number of operational cancellations where (STD-3h) ≤ ATOC < STD number of operational cancellations where ATOC ≥ STD
Other Checks	
Notes	If the actual time of cancellation is not available, the value 01-01-1990 00:00:00 shall be used instead.

[1] AODB: Airport Operational Database
[2] RMS: Resource Management System

5.21 ODI 21 Airport Departure Slot / Scheduled Off-Block Time (APDF-ODI-021-M)

ODI#	21
ID	SOBT
Version ID / Version date	00-11 / 04-09-2014
Item	Airport departure slot
Acronym(s) [Field Header(s)]	SOBT_UTC / SOBT_LT
IR390/13 Annex V ODI	3.2.2.i and 3.2.3.f
IR390/13 Annex V Definition	3.1.q 'airport arrival slot' and 'airport departure slot' mean an airport slot assigned either to an arrival or departure flight as defined in Regulation (EEC) No 95/93
Alternative definition	
Source	
Background info	Regulation (EEC) No 95/93
Scope	departures and each arrival operational cancellations where a departure airport slot was allocated
Data format	
Data type	DateTime DD-MM-YYYY hh:nn:ss
Data note	time in UTC or LT (Local Time) is sufficient but both may be provided.
Permissible value domain	
Example	11-01-2008 09:05:00
Quality Checks	
Relevance	key item for (K)PIs calculation
Accuracy	
Coherence and Comparability	% Completeness check for flights on NM database ≥ 98% of completeness for IFR flights (flight rules I, Y or Z)
Other Checks	
Notes	Mandatory for level 3 airports Recommended for level 2 and 1 airports based on facilitated schedules

5.22 ODI 22 Airport Arrival Slot / Scheduled In-Block Time (APDF-ODI-022-M)

ODI#	22
ID	SIBT
Version ID / Version date	00-11 / 04-09-2014
Item	Airport arrival slot
Acronym(s) [Field Header(s)]	SIBT_UTC / SIBT_LT
IR390/13 Annex V ODI	
IR390/13 Annex V Definition	3.2.2.i and 3.2.3.f
Alternative definition	3.1.q 'airport arrival slot' and 'airport departure slot' mean an airport slot assigned either to an arrival or departure flight as defined in Regulation (EEC) No 95/93
Source	
Background info	Regulation (EEC) No 95/93
Scope	arrivals and each arrival operational cancellation where an airport slot was allocated
Data format	
Data type	DateTime DD-MM-YYYY hh:nn:ss
Data note	time in UTC or LT (Local Time) is sufficient but both may be provided.
Permissible value domain	
Example	11-01-2008 09:05:00
Quality Checks	
Relevance	
Accuracy	
Coherence and Comparability	% Completeness check for flights on NM database ≥ 98% of completeness for IFR flights (flight rules I, Y or Z)
Other Checks	
Notes	Mandatory for level 3 airports Recommended for level 2 and 1 airports based on facilitated schedules

5.23 ODI 23 IATA Service Type (APDF-ODI-023-M)

ODI#	23
ID	SVCTYP
Version ID / Version date	00-11 / 04-09-2014
Item	IATA Service Type
Acronym(s) [Field Header(s)]	SVCTYP
IR390/13 Annex V ODI	
IR390/13 Annex V Definition	
Alternative definition	Code used for the classification of a flight or flight-leg as well as the type of service provided.
Source	IATA Standard Schedules Information Exchange – Appendix C Service Types
Background info	
Scope	highly desirable for arrivals, departures and operational cancellations
Data format	
Data type	alphanumeric string
Data note	
Permissible value domain	[A-X]{1}
Example	J
Quality Checks	
Relevance	
Accuracy	
Coherence and Comparability	
Other Checks	
Notes	

5.24 ODI 24 IFPS Flight Plan ID (APDF-ODI-024-M)

ODI#	24
ID	IFPLID
Version ID / Version date	00-11 / 04-09-2014
Item	IFPS Flight Plan ID
Acronym(s) [Field Header(s)]	IFPLID
IR390/13 Annex V ODI	
IR390/13 Annex V Definition	
Alternative definition	Unique flight plan identifier, assigned by the IFPS.
Source	EUROCONTROL Specification for ATS Data Exchange Presentation (ADEXP)
Background info	
Scope	highly desirable for arrivals and departures, and operational cancellations
Data format	
Data type	alphanumeric string
Data note	
Permissible value domain	[A-Z]{2}[0-9]{8}
Example	AA12345678
Quality Checks	
Relevance	key association field
Accuracy	
Coherence and Comparability	
Other Checks	
Notes	Flight identification shall be mainly based upon the IFPLID if available. The REG, ARCTYP, FLTID, ADEP, ADES and OOOI timestamps are included in the matching methods for readability and cross-check purposes.

6. Annexes

6.1 Standard IATA Delay Codes (AHM 730 & 731)

	Airline Internal Codes
00-05	AIRLINE INTERNAL CODES
	Others
06 (OA)	NO GATE/STAND AVAILABILITY DUE TO OWN AIRLINE ACTIVITY
09 (SG)	SCHEDULED GROUND TIME LESS THAN DECLARED MINIMUM GROUND TIME
	Passenger and Baggage
11 (PD)	LATE CHECK-IN, acceptance after deadline
12 (PL)	LATE CHECK-IN, congestions in check-in area
13 (PE)	CHECK-IN ERROR, passenger and baggage
14 (PO)	OVERSALES, booking errors
15 (PH)	BOARDING, discrepancies and paging, missing checked-in passenger
16 (PS)	COMMERCIAL PUBLICITY/PASSENGER CONVENIENCE, VIP, press, ground meals and missing personal items
17 (PC)	CATERING ORDER, late or incorrect order given to supplier
18 (PB)	BAGGAGE PROCESSING, sorting etc.
19 (PW)	REDUCED MOBILITY, boarding / deboarding of passengers with reduced mobility.
	Cargo and Mail
21 (CD)	DOCUMENTATION, errors etc.
22 (CP)	LATE POSITIONING
23 (CC)	LATE ACCEPTANCE
24 (CI)	INADEQUATE PACKING
25 (CO)	OVERSALES, booking errors
26 (CU)	LATE PREPARATION IN WAREHOUSE
27 (CE)	DOCUMENTATION, PACKING, etc. (Mail Only)
28 (CL)	LATE POSITIONING (Mail Only)
29 (CA)	LATE ACCEPTANCE (Mail Only)
	Aircraft and Ramp Handling
31 (GD)	AIRCRAFT DOCUMENTATION LATE/INACCURATE, weight and balance, general declaration, pax manifest, etc.
32 (GL)	LOADING/UNLOADING, bulky, special load, cabin load, lack of loading staff
33 (GE)	LOADING EQUIPMENT, lack of or breakdown, e.g. container pallet loader, lack of staff
34 (GS)	SERVICING EQUIPMENT, lack of or breakdown, lack of staff, e.g. steps
35 (GC)	AIRCRAFT CLEANING
36 (GF)	FUELLING/DEFUELLING, fuel supplier
37 (GB)	CATERING, late delivery or loading
38 (GU)	ULD, lack of or serviceability
39 (GT)	TECHNICAL EQUIPMENT, lack of or breakdown, lack of staff, e.g. pushback
	Technical and Aircraft Equipment
41 (TD)	AIRCRAFT DEFECTS.
42 (TM)	SCHEDULED MAINTENANCE, late release.
43 (TN)	NON-SCHEDULED MAINTENANCE, special checks and/or additional works beyond normal maintenance schedule.
44 (TS)	SPARES AND MAINTENANCE EQUIPMENT, lack of or breakdown.
45 (TA)	AOG SPARES, to be carried to another station.
46 (TC)	AIRCRAFT CHANGE, for technical reasons.
47 (TL)	STAND-BY AIRCRAFT, lack of planned stand-by aircraft for technical reasons.
48 (TV)	SCHEDULED CABIN CONFIGURATION/VERSION ADJUSTMENTS.
	Damage to Aircraft & EDP/Automated Equipment Failure
51 (DF)	DAMAGE DURING FLIGHT OPERATIONS, bird or lightning strike, turbulence, heavy or overweight landing, collision during taxiing
52 (DG)	DAMAGE DURING GROUND OPERATIONS, collisions (other than during taxiing), loading/off-loading damage, contamination, towing, extreme weather conditions
55 (ED)	DEPARTURE CONTROL
56 (EC)	CARGO PREPARATION/DOCUMENTATION
57 (EF)	FLIGHT PLANS
58 (EO)	OTHER AUTOMATED SYSTEM

Flight Operations and Crewing

61 (FP) FLIGHT PLAN, late completion or change of, flight documentation
62 (FF) OPERATIONAL REQUIREMENTS, fuel, load alteration
63 (FT) LATE CREW BOARDING OR DEPARTURE PROCEDURES, other than connection and standby (flight deck or entire crew)
64 (FS) FLIGHT DECK CREW SHORTAGE, sickness, awaiting standby, flight time limitations, crew meals, valid visa, health documents, etc.
65 (FR) FLIGHT DECK CREW SPECIAL REQUEST, not within operational requirements
66 (FL) LATE CABIN CREW BOARDING OR DEPARTURE PROCEDURES, other than connection and standby
67 (FC) CABIN CREW SHORTAGE, sickness, awaiting standby, flight time limitations, crew meals, valid visa, health documents, etc.
68 (FA) CABIN CREW ERROR OR SPECIAL REQUEST, not within operational requirements
69 (FB) CAPTAIN REQUEST FOR SECURITY CHECK, extraordinary

Weather

71 (WO) DEPARTURE STATION
72 (WT) DESTINATION STATION
73 (WR) EN ROUTE OR ALTERNATE
75 (WI) DE-ICING OF AIRCRAFT, removal of ice and/or snow, frost prevention excluding unserviceability of equipment
76 (WS) REMOVAL OF SNOW, ICE, WATER AND SAND FROM AIRPORT
77 (WG) GROUND HANDLING IMPAIRED BY ADVERSE WEATHER CONDITIONS

Air Traffic Flow Management Restrictions

81 (AT) ATFM due to ATC EN-ROUTE DEMAND/CAPACITY, standard demand/capacity problems
82 (AX) ATFM due to ATC STAFF/EQUIPMENT EN-ROUTE, reduced capacity caused by industrial action or staff shortage, equipment failure, military exercise or extraordinary demand due to capacity reduction in neighbouring area
83 (AE) ATFM due to RESTRICTION AT DESTINATION AIRPORT, airport and/or runway closed due to obstruction, industrial action, staff shortage, political unrest, noise abatement, night curfew, special flights
84 (AW) ATFM due to WEATHER AT DESTINATION

Airport and Governmental Authorities

85 (AS) MANDATORY SECURITY
86 (AG) IMMIGRATION, CUSTOMS, HEALTH
87 (AF) AIRPORT FACILITIES, parking stands, ramp congestion, lighting, buildings, gate limitations, etc.
88 (AD) RESTRICTIONS AT AIRPORT OF DESTINATION, airport and/or runway closed due to obstruction, industrial action, staff shortage, political unrest, noise abatement, night curfew, special flights
89 (AM) RESTRICTIONS AT AIRPORT OF DEPARTURE WITH OR WITHOUT ATFM RESTRICTIONS, including Air Traffic Services, start-up and pushback, airport and/or runway closed due to obstruction or weather1, industrial action, staff shortage, political unrest, noise abatement, night curfew, special flights

Reactionary

91 (RL) LOAD CONNECTION, awaiting load from another flight
92 (RT) THROUGH CHECK-IN ERROR, passenger and baggage
93 (RA) AIRCRAFT ROTATION, late arrival of aircraft from another flight or previous sector
94 (RS) CABIN CREW ROTATION, awaiting cabin crew from another flight
95 (RC) CREW ROTATION, awaiting crew from another flight (flight deck or entire crew)
96 (RO) OPERATIONS CONTROL, re-routing, diversion, consolidation, aircraft change for reasons other than technical

Miscellaneous

97 (MI) INDUSTRIAL ACTION WITH OWN AIRLINE
98 (MO) INDUSTRIAL ACTION OUTSIDE OWN AIRLINE, excluding ATS
99 (MX) OTHER REASON, not matching any code above

SOURCE: IATA – Airport Handling Manual (AHM730)

6.2 Standard IATA Delay Sub-Codes (AHM 731)

73 (WR) **WEATHER: EN ROUTE OR ALTERNATE**
- **Z** OUTSIDE AIRCRAFT LIMITS
- **Y** OUTSIDE CREW LIMITS
- **X** ETOPS

81 (AT) **ATFM DUE TO ATC EN-ROUTE DEMAND/CAPACITY, standard demand/capacity problems**
- **Z** ATC ROUTEINGS
- **Y** HIGH DEMAND OR CAPACITY

82 (AX) **ATFM DUE TO ATC STAFF/EQUIPMENT EN-ROUTE, reduced capacity caused by industrial action or shortage or equipment failure, extraordinary demand due to capacity reduction in neighbouring area**
- **Z** INDUSTRIAL ACTION
- **Y** EQUIPMENT FAILURE
- **X** STAFF SHORTAGE
- **W** MILITARY ACTIVITY
- **V** SPECIAL EVENT

83 (AE) **ATFM DUE TO RESTRICTION AT DESTINATION AIRPORT, airport and/or runway closed due to obstruction, industrial action, staff shortage, political unrest, noise abatement, night curfew, special flights**
- **Z** HIGH DEMAND / ATC CAPACITY
- **Y** INDUSTRIAL ACTION
- **X** EQUIPMENT FAILURE
- **W** STAFF SHORTAGE
- **V** ACCIDENT / INCIDENT
- **U** MILITARY ACTIVITY
- **T** SPECIAL EVENT
- **S** NOISE ABATEMENT/NIGHT CURFEW
- **R** OTHER

87 (AF) **AIRPORT FACILITIES, parking stands, ramp congestion, lighting, buildings, gate limitations, etc.**
- **Z** LACK OF PARKING STANDS
- **Y** RAMP CONGESTION
- **X** LIGHTING OR BUILDINGS
- **W** GATE LIMITATION / NO GATE AVAILABLE
- **V** BAGGAGE SORTING SYSTEM DOWN / SLOW
- **U** NO PUSH BACK CLEARANCE DUE TO INFRASTRUCTURE
- **T** JET BRIDGE INOPERATIVE
- **S** LACK OF CHECK IN COUNTERS
- **R** ELECTRICAL SYSTEM FAILURE
- **P** PASSENGER TRANSPORT SYSTEM FAILURE
- **N** PUBLIC ADDRESS/FLIGHT INFORMATION DISPLAY SYSTEM FAILURE
- **M** INSUFFICIENT FIRE COVER
- **L** GROUND COMMUNICATION SYSTEM FAILURE
- **K** NO PUSH BACK CLEARANCE DUE TO CONSTRUCTION
- **J** BREAKDOWN OF AIRPORT FUELLING SYSTEM
- **H** LATE OR LACK OF FOLLOW ME FOR PUSH-BACK
- **G** ANY OF THE ABOVE AT THE DESTINATION AIRPORT

89 (AM) **RESTRICTIONS AT AIRPORT OF DEPARTURE WITH OR WITHOUT ATFM RESTRICTIONS, including Air Traffic Services, start-up and pushback, airport and/or runway closed due to obstruction or weather.**
- **Z** ATC CAPACITY
- **Y** ATC INDUSTRIAL ACTION
- **X** ATC STAFFING
- **W** ATC EQUIPMENT
- **V** ATC ACCIDENT/INCIDENT
- **U** ATC DUE TO DE-ICING
- **T** ATC SPECIAL EVENT
- **S** ATC WEATHER
- **R** ATC RESTRICTIONS DUE TO CURFEW
- **P** ATC POLITICAL UNREST
- **N** ATC ENVIRONMENTAL
- **M** AIRPORT CLOSURE
- **L** RUNWAY CLOSURE
- **K** START-UP/PUSHBACK CLEARENCE DELAY (LOCAL ATC)
- **J** LOST FLIGHT PLAN BY ATC
- **H** CONSTRUCTION WORK/MAINTENENCE
- **G** OTHER

93 (RA) **AIRCRAFT ROTATION, late arrival of aircraft from another flight**
- **Z** LATE ARRIVAL DUE DEPARTURE DELAY AT PREVIOUS STATION
- **Y** LATE ARRIVAL DUE ENROUTE DELAY
- **X** LATE ARRIVAL DUE DELAY AFTER LANDING
- **W** LATE ARRIVAL DUE TO HIGH DEMAND FOR DESTINATION STATION
- **V** LATE ARRIVAL DUE TO WEATHER AT DESTINATION
- **U** LATE ARRIVAL DUE TO TECHNICAL REASONS

SOURCE: IATA – Airport Handling Manual (AHM731)

6.3 Standard IATA Change Reasons (SSIM Chapter 2)

AIRS	Airspace restrictions
ARPT	Airfield restrictions
COMM	Commercial reasons demand or lack of demand
CREW	Crew shortage
DAMA	Aircraft damage
EQUI	Equipment shortage
FUEL	Fuel shortage
HDLG	Ground handling
HOLI	Holiday
INDU	Industrial dispute
OPER	Operational reasons
PERF	Aircraft performance
POLI	Political situation
POSI	Aircraft positioning
REPO	Aircraft re-positioning
ROTA	Aircraft rotation
RTNS	Return to normal schedule or reinstatement of flight status prior to issuance of ASM(s) (withdrawal of ASM change)
RUNW	Runway restrictions
TECH	Technical reasons, maintenance, etc.
WEAT	Weather conditions

SOURCE: IATA – Standard Schedules Information Manual

6.4 IATA Service Types (SSIM Appendix C)

Service Type Code	Application	Type of Operation	Description
J	Scheduled	Passenger	Normal Service
S	Scheduled	Passenger	Shuttle Mode
U	Scheduled	Passenger	Service operated by Surface Vehicle
F	Scheduled	Cargo/Mail	Loose Loaded cargo and/or preloaded devices
V	Scheduled	Cargo/Mail	Service operated by Surface Vehicle
M	Scheduled	Cargo/Mail	Mail only
Q	Scheduled	Passenger/Cargo	Passenger/Cargo in Cabin (mixed configuration aircraft)
G	Additional Flights	Passenger	Normal Service
B	Additional Flights	Passenger	Shuttle Mode
A	Additional Flights	Cargo/Mail	Cargo/Mail
R	Additional Flights	Passenger/Cargo	Passenger/Cargo in Cabin (mixed configuration aircraft)
C	Charter	Passenger	Passenger Only
O	Charter	Special Handling	Charter requiring special handling (e.g. Migrants/immigrants flights)
H	Charter	Cargo/Mail	Cargo and/or Mail
L	Charter	Passenger/Cargo/Mail	Passenger and Cargo and/or Mail
P	Others	Not specific	Non-revenue (Positioning/Ferry/Delivery/Demo)
T	Others	Not specific	Technical test
K	Others	Not specific	Training (School/Crew check)
D	Others	Not specific	General Aviation
E	Others	Not specific	Special (FAA/Government)
W	Others	Not specific	Military
X	Others	Not specific	Technical Stop
I	Others	Not specific	State/Diplomatic/Air Ambulance
N	Others	Not specific	Business Aviation/Air taxi

SOURCE: IATA – Standard Schedules Information Manual

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Revision History

Edition	Description	Comment
00-01	new draft – all pages	new document
00-02	establishment of normative elements of data specification	
00-03	inclusion of internal review comments	
00-04	introduction of requirements numbering convention	
00-05	refinements based on technical interface testing	
00-06	inclusion of operational data item specification	
00-07	update of operational data item specification	
00-08	alignment with data policies and requirement numbering conventions	
00-09	finalisation of draft for stakeholder consultation and roll-out	technical readiness testing completed and consolidated for this release
00-10	refinement and amendments based on stakeholder consultation during RP2 preparatory action and technical interface testing	release candidate of specification
00-11	errata amendment	

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